

Graph RAG for Legal Norms: A Hierarchical and Temporal Approach

Hudson de Martim¹

¹Federal Senate of Brazil , hudsonm@senado.leg.br

Abstract

This article proposes an adaptation of Graph Retrieval Augmented Generation (Graph RAG) specifically designed for the analysis and comprehension of legal norms, which are characterized by their predefined hierarchical structure, extensive network of internal and external references and multiple temporal versions. By combining structured knowledge graphs with contextually enriched text segments, Graph RAG offers a promising solution to address the inherent complexity and vast volume of legal data. The integration of hierarchical structure and temporal evolution into knowledge graphs - along with the concept of comprehensive Text Units - facilitates the construction of richer, interconnected representations of legal knowledge. Through a detailed analysis of Graph RAG and its application to legal norm datasets, this article aims to advance the field of Artificial Intelligence applied to Law, creating opportunities for more effective systems in legal research, legislative analysis, and decision support.

Keywords: Retrieval-Augmented Generation; Knowledge Graphs; Legal Tech; Legal AI; Legal Information Retrieval; Knowledge Representation.

1 Introduction

In the contemporary era, where dependence on precise and accessible information is growing, the legal domain presents a unique challenge to professionals and analysis systems due to its vast volume and intrinsic complexity. Navigating this mass of data and efficiently extracting relevant information becomes a progressively arduous task. This difficulty not only burdens legal professionals but also exposes the limitations of conventional review and analysis methods – whether manual or digital – which are often slow and susceptible to errors. This complexity is aggravated by legal language itself, which, with its specific terminology and requirement for deep contextual understanding, presents essential nuances for rigorous interpretation and analysis.

The challenge is amplified by three defining characteristics of legal norms: the predefined hierarchical structure, the extensive cross-references (internal and external) and the multiplicity of temporal versions. In this scenario, Artificial Intelligence (AI) emerges as a promising field, offering tools and techniques capable of overcoming the barriers of traditional legal analysis and optimizing access to and comprehension of this vast knowledge.

Legal frameworks are inherently hierarchical. This structure manifests both between different types of norms (constitutions, statutes, regulations) and within individual legal texts themselves. Higher-level legal principles (e.g., constitutional provisions) frequently influence or restrict lower-level norms (e.g., specific regulations) – regarding the hierarchy of norms and legal principles, see [1]. On the other hand, legal norms are typically organized internally through grouping provisions such as titles, chapters, sections, and further subdivided into specific provisions like articles, paragraphs, items, clauses etc. Understanding

the relationships across all levels of this hierarchy – both inter-norm and intra-norm – is essential for the interpretation and application of Law.

Additionally, the dynamic nature of legal norms, evolving through amendments, repeals, and new interpretations, results in multiple temporal versions. The ability to track and understand this evolution [2] is, therefore, essential for determining the applicable legislation at a given moment.

These defining characteristics of legal norms, although representing significant analytical challenges, are indispensable for correct application and normative interpretation. Consequently, they demand analysis approaches that explicitly capture the structural and temporal dimensions of the legal norms, precisely the objective of the adapted Graph-based Retrieval-Augmented Generation (Graph RAG) methodology in this article.

Graph RAG emerges as an evolution of traditional RAG [3], which combines information retrieval with generation by Large Language Models (LLMs)¹ to enhance the relevance and precision of content. By enabling LLMs to base their answers on external knowledge sources, going beyond the data used in their training, RAG mitigates inherent limitations of these models, such as the propensity for hallucinations².

By utilizing knowledge graphs to represent and structure the relationships between concepts and entities extracted from these knowledge sources, Graph RAG [4] goes beyond traditional RAG by capturing the intricate and interconnected relationships within and between documents, which is particularly valuable for complex domains like Law. Legal knowledge is highly relational. Statutes refer to other statutes, clauses reference other clauses, and jurisprudence is based on precedents. Knowledge graphs are well-suited to represent these complex relationships.

Graph RAG presents significant potential for the analysis of legal norms. This technique can integrate structured information from knowledge graphs with unstructured textual content to create richer and more interconnected representations of legal knowledge. This integration allows capturing the complex relationships between legal entities, such as laws, doctrine, jurisprudence, concepts, and the temporal evolution of legal texts. By representing these legal entities, and their relationships, in a graph and linking them to the actual text of the legal norms and their different versions, Graph RAG can provide a more holistic understanding of Law [5].

The objective of this article is to contribute to the advancement of knowledge in Artificial Intelligence applied to Law, detailing the analysis of Graph RAG and its application to datasets of legal norms, with the goal of facilitating more effective systems to support legislative and legal work.

This paper is structured as follows. Section 2 first defines the core structural entities – Norm, Component, and Version – essential for modeling legal documents, discussing their hierarchical relationships and the challenges of representing temporal changes. Section 3 then elaborates on our proposed adaptations to the Graph RAG framework, introducing the concept of Text Units, explaining the rationale for modeling versions as aggregations rather than simple compositions, detailing the inclusion of essential metadata and alteration actions, and discussing strategies for structure-aware retrieval. Lastly, Section 4 concludes the paper by highlighting the key contributions and suggesting avenues for future investigation.

2 Structural Entities in the Graph

A Graph RAG system operates through the combination of several essential components. At its core lies a knowledge graph [6], which serves as a structured representation of knowledge. In this graph, nodes represent entities such as legal concepts, articles of law, or judicial cases, while edges symbolize the relationships

¹LLMs (Large Language Models), such as GPT-4 or Gemini 2.0, are AI models trained to understand and generate natural language. For legal applications, see [16].

²LLMs are trained on vast amounts of data but may lack specific or updated information. RAG addresses this problem by retrieving relevant documents and providing them as context to the LLM before it generates a response.

between these entities, such as definitions, citations, or modifications. The knowledge graph provides a semantic foundation for understanding these connections. It explicitly models relationships, which are often implicit in unstructured text, making them accessible for retrieval and reasoning.

The initial step in both Graph RAG and traditional RAG approaches is to divide each text in the input dataset into segments, or “*text chunks*” - these segments are, throughout the process, converted into embeddings [7] and stored in a vector database [8]. Various segmentation strategies can be employed, such as sentence segmentation, paragraph segmentation, window-based splitting (according to the embedding model’s window size), and semantic segmentation (dividing text according to its semantic blocks), among others.

The second step in Graph RAG involves extracting *Entities*, along with their properties and relationships, from each text segment generated in the previous step. These extracted entities serve as the “building blocks” or nodes of the knowledge graph, with edges interconnecting them to represent the relationships derived from the normative texts.

In the context of legal norms, the entities extracted from the texts can be categorized into two groups:

1. **Content Entities:** traditional entities in the text of a legal norm such as types *Person*, *Organization*, *Location*, *Event*, *Concept* (metamodel entity³), *Legal Norm* (normative reference⁴) etc.
2. **Structural Entities**
 - entities that represent the *hierarchical structure* of the text of a legal norm (its *articulation*), according to the formation rules of this type of text;
 - entities that represent the *versions* of this structure over time.

2.1 Legal Norm Structure

To structure the text of a Brazilian legal norm [9], the following components are defined⁵:

- **da Parte Inicial** (of the Initial Part):
 - **Fórmula de Promulgação** (Promulgation Formula): indicates the authority or institution responsible for creating the norm.
 - **Epígrafe** (Epigraph): presents the full name of the norm.
 - **Ementa** (Abstract/Summary): provides a concise summary of the norm’s content.
 - **Preâmbulo** (Preamble): offers an introductory text that indicates the proposing body of the norm and, when applicable, its legal basis.
- **Articulação** (Articulation):
 - Encompasses the legal provisions, structured into **Dispositivos**⁶ (provisions) as illustrated in figures 1 and 2.

³Legal metamodel, in this context, refers to a conceptual model that defines the fundamental types of entities and relationships in the domain of Law, such as abstract legal concepts, principles, values, etc. Examples would be: "Right to Life", "Principle of Legality", "Social Contract".

⁴Normative reference represents a remission or citation to another legal norm, either in its entirety or to a specific provision.

⁵In a more precise representation of the structuring of a legal norm, there would be the element **Componente Autônomo Articulado** (Autonomous Articulated Component), which would group all parts of the norm, including annexes. But, for didactic simplification, it will not be presented.

⁶A **Dispositivo** (provision) of a legal norm is each element/component of the text that contains the prescription, i.e., the command, the rule of conduct, the permission, the prohibition, or the authorization that the norm establishes. It forms the normative core of the legal text.

- **da Parte Final** (of the Final Part):

- **Fecho** (Closing): details the place and date of issuance of the norm.
- **Assinaturas e Assinaturas Complementares** (Signatures and Complementary Signatures): include the necessary signatory information.

This hierarchical organization not only facilitates the systematic interpretation of legal texts but also enables precise navigation and retrieval of specific legal provisions.

The basic unit of a Brazilian legal norm's articulation is the **Artigo** (Article). Each Article can be further subdivided into additional provisions, as demonstrated in Figure 1.

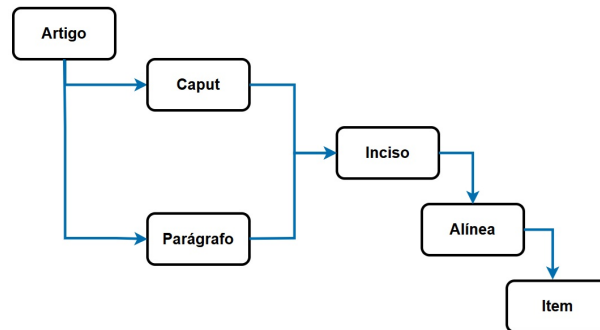


Figure 1: Diagram of the hierarchical structure of possible subdivisions for the Article (Artigo) provision in Brazilian legislation. It shows Artigo (Article) → Caput (main section); Artigo (Article) → Parágrafo (Paragraph); Caput, Parágrafo (Paragraph) → Inciso (Item); Inciso (Item) → Alínea (Subitem); Alínea (Subitem) → Item (Sub-subitem).

Articles can also be organized into **Dispositivos de Agrupamento** (Grouping Provisions), as illustrated in Figure 2.

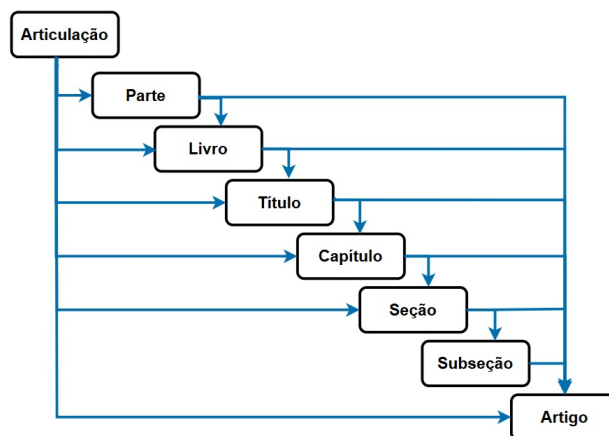


Figure 2: Diagram of the hierarchical structure of grouping for the "Article" (Artigo) provision in Brazilian legislation. It shows Articulação (Articulation) → Parte (Part) → Livro (Book) → Título (Title) → Capítulo (Chapter) → Seção (Section) → Subseção (Subsection) → Artigo (Article).

As an example, Figure 3 presents the provision “Art. 12” of the “Constituição Federal Brasileira de 1988” (Federal Constitution of Brazil (1988)), displaying its grouping provisions (“Título II” and “Capítulo III”)

and some of its child provisions (or subdivisions). Highlighted is the article, the basic unit of articulation. Annotations in orange are representing the type of each provision.

TÍTULO II DOS DIREITOS E GARANTIAS FUNDAMENTAIS [Título (Title)]
 CAPÍTULO III DA NACIONALIDADE [Capítulo (Chapter)]
 Art. 12. [Artigo (Article)]
 São brasileiros: [Caput]
 I – natos: [Inciso (Item)]
 a) os nascidos na República Federativa do Brasil, ainda que de pais estrangeiros, desde que estes não estejam a serviço de seu país; [Alínea (Subitem)]
 b) os nascidos no estrangeiro, de pai brasileiro ou de mãe brasileira, desde que qualquer deles esteja a serviço da República Federativa do Brasil; [Alínea (Subitem)]
 c) os nascidos no estrangeiro de pai brasileiro ou de mãe brasileira, desde que sejam registrados em repartição brasileira competente ou venham a residir na República Federativa do Brasil e optem, em qualquer tempo, depois de atingida a maioridade, pela nacionalidade brasileira; [Alínea (Subitem)]
 II – naturalizados: [Inciso (Item)]
 a) os que, na forma da lei, adquiram a nacionalidade brasileira, exigidas aos originários de países de língua portuguesa apenas residência por um ano ininterrupto e idoneidade moral; [Alínea (Subitem)]
 b) os estrangeiros de qualquer nacionalidade, residentes na República Federativa do Brasil há mais de quinze anos ininterruptos e sem condenação penal, desde que requeiram a nacionalidade brasileira. [Alínea (Subitem)]

Figure 3: Example of articulated text for Art. 12 of the Federal Constitution of Brazil (1988) with annotations indicating the types of hierarchical provisions

As proposed, there will be 3 (three) types of *Structural Entities* for building the Knowledge Graph:

1. **Norm (as a Whole):** represents a legal norm.
2. **Component:** represents an element of the hierarchical structure of a legal norm. It can optionally be composed of other components (its subdivisions).
3. **Version:** version of the content (e.g., label and text) of a component, or of a norm (as a whole), over time – will be detailed further ahead.

The *Norm* and *Component* types are found in the **Hierarchy dimension**. The *Version* type is in the **Time dimension** [10].

2.2 Legal Norm Text Segmentation

In the original Graph RAG proposal, the first step involves segmenting the entire document into smaller text segments, or “text chunks”. In the subsequent step, each segment is processed—typically using a large language model (LLM)—to extract the identified entities, along with their properties and relationships, which are then added to the Knowledge Graph. However, when dealing with legal texts, the hierarchical structure is a fundamental characteristic that must be preserved to ensure accurate interpretation. Therefore, it is essential to extract and represent this hierarchical structure within the Knowledge Graph by mapping it to the previously defined structural entities.

Ideally, semantic segmentation⁷ should be employed to divide the document into text segments that correspond directly to its structural elements, as illustrated in Figure 3. In this approach, the initial segmentation

⁷Semantic segmentation is a division of the document based on the meaning and structure of the text, rather than merely on superficial textual criteria such as sentences or paragraphs.

step would simultaneously extract the structural entities, thereby capturing the hierarchical organization from the outset - the subsequent step would focus on extracting content entities, as outlined in the original Graph RAG proposal.

This proposed new first step (semantic segmentation oriented by the hierarchical structure of the legal norm text) could be done using an LLM (with fine-tuning or prompt engineering) prepared to recognize the structure of legal norms in the worked domain, generating a structured output in XML or JSON, for example. Another option would be to use a specialized Parser⁸ for segments of legal norm texts that recognizes and processes this structure – in both options, human review is recommended, given the complexity and importance of precision in interpreting the legal structure, and to ensure the correctness and quality of segmentation and extraction.

As an example, a passage of the the “Constituição Federal Brasileira de 1988” (Federal Constitution of Brazil (1988)) received the semantic segmentation (like showed in Figure 3), producing the entity of type *Norm*, represented by a circle in green, and those of type *Component*, represented in blue in the hierarchy presented in Figure 4.

For a more precise representation in the Graph, specialized entity types, derived from *Norm*, can be created for each type of norm in the treated legal system. In Brazil, there are, for example: "Constituição Federal" (Federal Constitution), “Lei Ordinária” (Ordinary Law) etc. Thus, the entity representing a legal norm could be included in the graph using the specialized type.

Specialized entity types, derived from *Component*, can also be created for each type of provision found in the treated legal system. For example, in the Brazilian case, there are types like "Artigo" (Article), "Inciso" (Item), “Alínea” (Subitem), “Título” (Title), “Capítulo” (Chapter), presented in figures 1 and 2. Thus, the entity representing a component could be included in the graph using the specialized type.

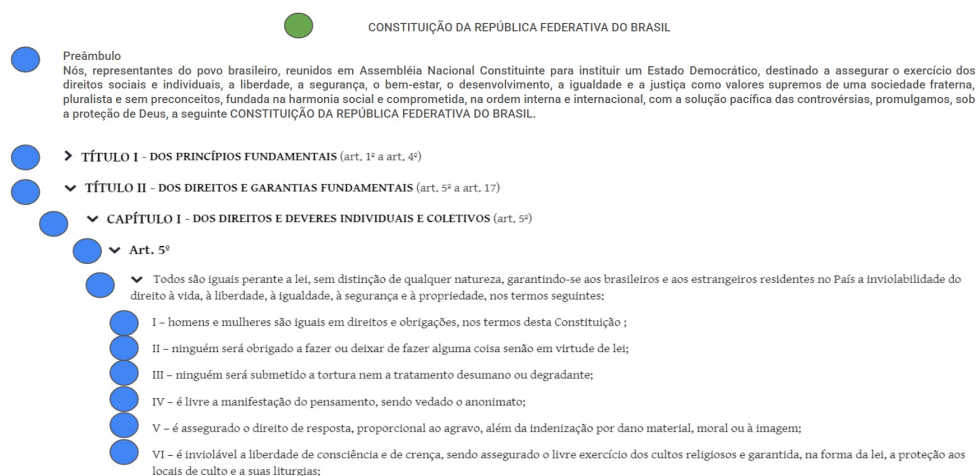


Figure 4: Illustration of hierarchical semantic segmentation, and typification of the structural entities, applied to a passage of the Federal Constitution of Brazil (1988), representing the "Norm" (green), "Components" (blue) and their associated texts.

⁸In the Brazilian case, an implementation of a Parser is available in the LexML project, which receives an unstructured text of a Brazilian legal norm and produces an XML file with the structured text. This parser, developed specifically for Brazilian legislation, represents a valuable tool for applying the proposed structured segmentation, as it automates the recognition and extraction of the hierarchical structure of norms, facilitating the creation of the Knowledge Graph. LeXML Brazil, Legislative Integration Project. Available at: <https://projeto.lexml.gov.br>. Accessed on: 15 Mar 2025.

2.3 Text Versioning over Time

Version entities represent the view over time of a norm/component, which can be extracted from its original text, but also from the “*compilação multivigente*”⁹ (multi-temporal compilation) of its text.

A component always has an original version, which corresponds to the text, and label, of the component on the signature date of the original norm. It optionally has versions created by subsequent norms that commanded, for example, alteration of the provision to new content (e.g., alteration of the wording of an article by an amendment), or commanded a repeal (e.g., an article being explicitly repealed by a later law), or a suppression (e.g., an item being suppressed by a later norm), or a renumbering (e.g., an article being renumbered, label altered, by a later law) etc. of that component.

Since the text of a norm is constructed by hierarchically grouping the texts of its individual components, the creation of a new version for any component automatically generates a new version for the entire norm as well as for all grouping elements/components that include that component.

For example, Figure 5 illustrates the *Version* entities for the provision “Art. 6” (and its caput - main section) of ‘Constituição Federal brasileira’ (Brazilian Federal Constitution)”¹⁰, demonstrating how different versions reflect changes over time.

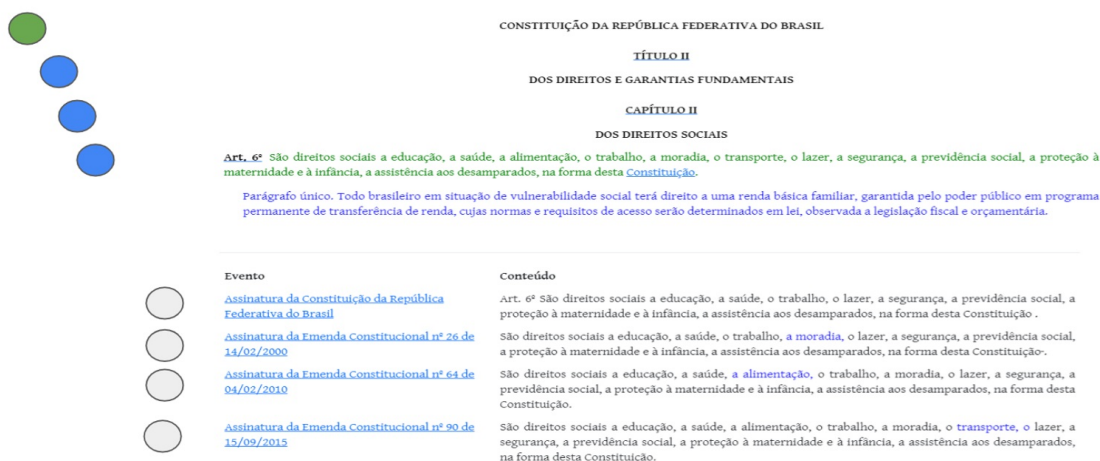


Figure 5: Structure of the provision "Art. 6" (and its caput - main section) of Brazilian Federal Constitution, illustrating four *Version* entities over time (light gray circles) and their respective generated events and resulting contents. The versions represent the different wordings of the caput on different dates, due to constitutional amendments.

2.4 Building the Graph

After semantically segmenting the text of a legal norm into a hierarchical structure (tree), as per the proposal presented above, the elements of the hierarchy are then inserted as nodes of the types *Norm* (with its associated properties, such as URN - Uniform Resource Name, name, signature date etc.) and *Component* (with its

⁹“Compilação” (Compilation) in Brazil is a process in which, starting from the original version of a legal norm, all modifications commanded by subsequent norms are applied to its text, up to the specified date, thus producing an unofficial updated text of this norm on this date. It is “Multivigente” (Multi-temporal) when all dates on which the norm underwent alteration are considered in the Compilation, producing an unofficial updated text (version) for each identified date.

¹⁰The structuring of the “Constituição Federal Brasileira de 1988” (Brazilian Federal Constitution of 1988) into “Norm”, “Components”, and “Versions” can be visualized at <https://normas.leg.br/?urn=urn:lex:br:federal:constituicao:1988-10-05;1988>. Accessed on 15.03.2025.

properties such as URN, optional inclusion date etc.) into the Knowledge Graph, as illustrated in Figure 4, which demonstrates the semantic segmentation of a Constitution passage.

Next, for each component, a node of the type *Version* will be inserted with its associated text segment [11] (and its properties such as URN, version date, label/name, text etc.) as presented in Figure 6 – this associated text segment (text chunk) is fundamental for the Retrieval Augmented Generation process, as it represents the textual content of that specific version that will be retrieved and used as context for the responses generated by the LLM.

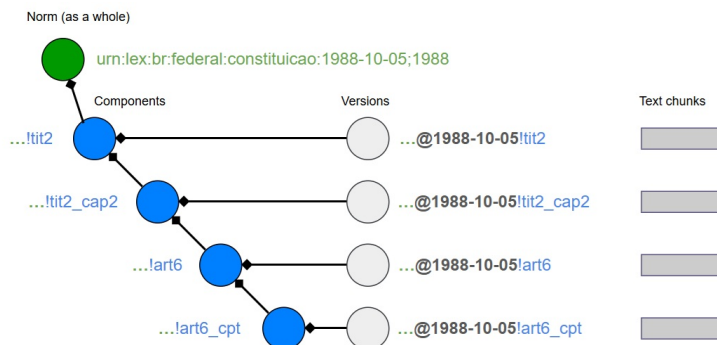


Figure 6: Representation in the knowledge graph of the relationship between hierarchical structure entities "Norm", "Component", their original versions "Version", and the associated Text Chunks.

For example, the *Norm* node/entity “Constituição Federal de 1988” (Federal Constitution of 1988) identified by urn¹¹

`urn:lex:br:federal:constituicao:1988-10-05;1988`

has a *Component* node/entity “caput do Art. 6º” (caput is the main text of the article) identified by

`urn:lex:br:federal:constituicao:1988-10-05;1988!art6_cpt`

which has associated the original version *Version* node/entity, identified by

`urn:lex:br:federal:constituicao:1988-10-05;1988@1988-10-05!art6_cpt`

that represents the version on the date 1988-10-05 (original version) of the text of the caput of Art. 6 of Federal Constitution of 1988. For this *Version* node/entity, we have a associated *Text chunk* with the text below - as illustrated in Figure 5:

"São direitos sociais a educação, a saúde, o trabalho, o lazer, a segurança, a previdência social, a proteção à maternidade e à infância, a assistência aos desamparados, na forma desta Constituição."

("Social rights include education, health, work, leisure, security, social security, protection of motherhood and childhood, and assistance to the destitute, as prescribed by this Constitution.")

Unlike other fields, where the text remains static over time (for example, the transcribed speech of a Parliamentarian in a Congress session will not change over time), the text over time of a legal norm, or of a component, can be represented by a set of Versions.

In Figure 7, four versions of the caput of Art. 6, found in Figure 5, are displayed: one original and three resulting from changes (text change) promoted by “Emendas Constitucionais” (Constitutional Amendments) - for a more precise representation in the Graph, specialized entity types, derived from *Version*, can be created for each type of version found in the treated legal system. For example, in Brazil, there are types like "Original Version", “Alteração de Texto” (Text Change Version), “Renumeração” (Renumbering Version), “Revogação” (Repeal Version) etc. Thus, the entity representing the version could be included in the graph using the specialized type.

¹¹The formation of the urns presented in this work follows the standard proposed by the Brazilian LexML project. Available at: <https://projeto.lexml.gov.br>. Accessed on: 15 Mar 2025.

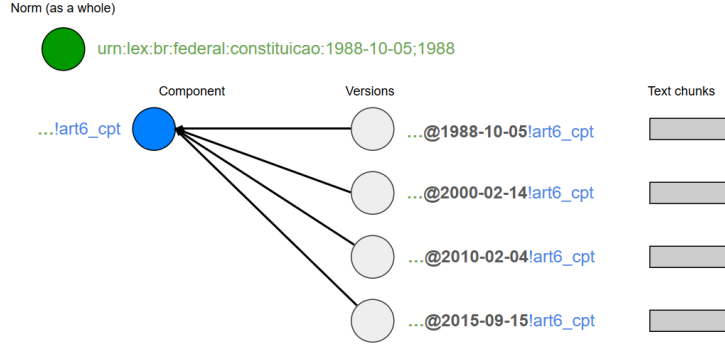


Figure 7: Representation of the multiple "Version" entities for the caput do Art. 6° provision (...!art6_cpt), illustrating its textual evolution over time and the link of each version to its Text Chunk.

The proposal of this work is not only to incorporate the entities of the hierarchical structure of the legal norm's text into the knowledge graph, but also to include the *Version* entities of each component of the norm. It is important to note that the text segments (text chunks), identified by the original text and by the versions of the text over time of the norm, will be linked to the *Version* entities, and not to *Component* entities, as they represent the content of the component at different moments in time, as shown in Figure 7 - in practice, this link could be materialized as a property of the *Version* entity in the graph, just like a property for the embedding of its content.

This temporal representation will allow answering questions about the text of a component considering the temporal context: for example, if there was any change in the text of a component over time; what changes the component underwent over time, or within a period; what was the text of the component on a specific date, etc.

3 Adapted approach

3.1 Text Units

Regarding the text segment (text chunk) associated with each *Version* entity, in cases of subdivisions of a caput, such as "*parágrafos*" (paragraphs) and enumeration elements ("*incisos*" (items) and "*alíneas*" (subitems)), we suggest adopting the strategy proposed by [12]. In this strategy, the text segment to be considered for a component is not only the text directly resulting from the semantic segmentation of the norm (or from its compilation over time) but the semantically complete text of the component, derived from post-segmentation (and post-compilation) processing. The cited author recognizes the hierarchical and interdependent nature of the components of a legal norm, where, for example, the meaning of an enumeration component (e.g., "*inciso*" (item)) is frequently dependent on the context of its parent component (e.g., the caput).

In other words, when including in the graph, for example, the original *Version* of the provision "inciso I do caput do Art. 5°" (item I of caput of Art. 5), presented in Figure 4, the text linked to this entity will not be just:

"I - homens e mulheres são iguais em direitos e obrigações, nos termos desta Constituição;"
 ("I - men and women are equal in rights and obligations, under the terms of this Constitution;")

but rather the text formed by the content of the article's caput (on the same date) plus the content of the item itself, which can also be visualized in Figure 8 - the text of the caput was placed as a prefix in square brackets.

"[Todos são iguais perante a lei, sem distinção de qualquer natureza, garantindo-se aos brasileiros e aos estrangeiros residentes no País a inviolabilidade do direito à vida, à liberdade, à igualdade, à segurança e à propriedade, nos termos seguintes:] I - homens e mulheres são iguais em direitos e obrigações, nos termos desta Constituição;"

("[Everyone is equal before the law, without distinction of any kind, guaranteeing to Brazilians and foreigners residing in the Country the inviolability of the right to life, liberty, equality, security, and property, under the following terms:] I – Men and women are equal in rights and obligations, under the terms of this Constitution;")

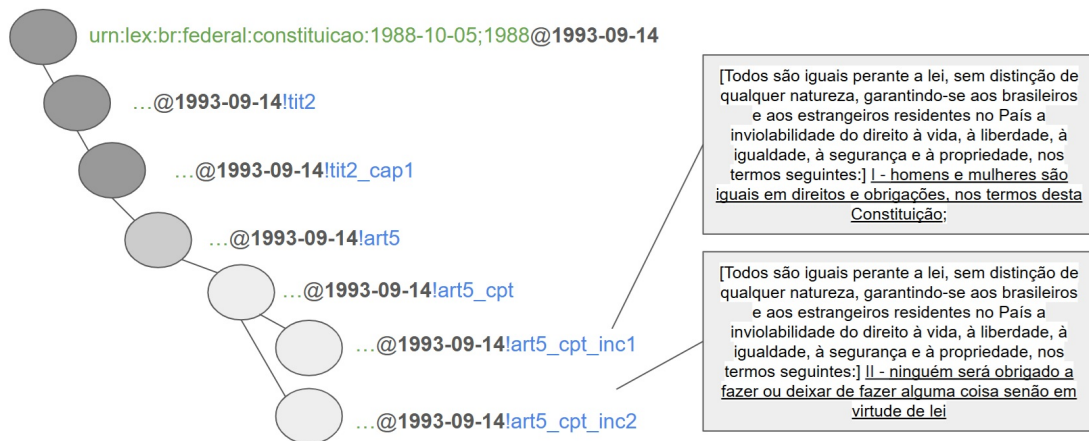


Figure 8: Example of contextual (cumulative) Text Units for "incisos" (items) of caput of Art. 5, demonstrating the inclusion of the parental caput text to provide complete semantic context.

The version of an article's caput will have as linked text its own text accumulated with the text from the tree of its child components, as exemplified in Figure 9. The same applies to the article, the groupers and the norm (as a whole), accumulating its own text with the text from their respective trees.

According to the cited author, this would bring more "aboutness"¹² to the texts linked to the entities [13]. In other words, the cumulative "Text Unit", by including the textual context of the parent, or child, components, represents the central theme and the legal meaning of the component in question more completely and comprehensively, increasing how well the text captures "what it is about" the provision.

Considering this new approach, we will call the texts associated with the entities *Text Units*, instead of Text Chunks, as they are not necessarily the pieces of text resulting directly from the segmentation (or compilation) of the norms' texts, but rather the contextually complete content of each component.

3.2 Versions as Aggregations

In addition to the original version of a component C of a legal norm N, for each date/time T on which a subsequent norm Y commands an alteration in the content of C, a new version T must be produced for component C, as presented in Figure 10.

Considering the proposal of Text Units resulting from the accumulation of text in the hierarchy, consequently, a new version T must also be produced for every component of norm N that:

1. groups, directly or indirectly, component C, including the norm (as a whole, as level 0);
2. has its Text Unit formed by the accumulation of the text of C as a prefix (for example: "parágrafos" (paragraphs), "incisos" (items) and "alíneas" (subitems)).

¹²The term "aboutness" refers to the capacity of a text to faithfully represent its central theme. For theoretical discussions, see [17].

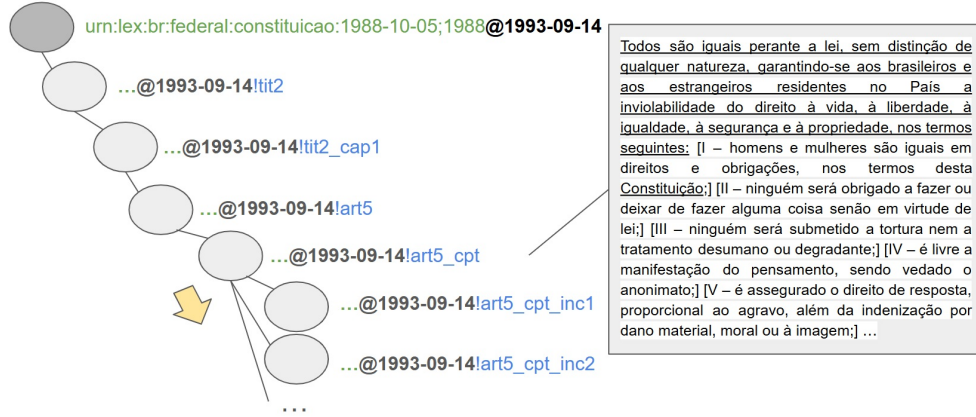


Figure 9: Example of cumulative Text Unit for the caput of Art. 5 of the "Constituição Federal" (Federal Constitution).

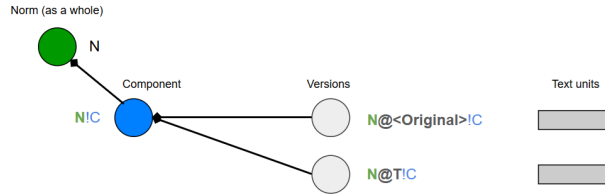


Figure 10: Versions of component C over time

In Figure 11, we observe that on the norm's original date (1988-10-05), the original version of the component "tit2" is formed by the original versions of its children. That is, the text of "Título II" (Title II) on that date consists of its own text and the texts of its children from that same date.

On 1993-09-14, a new version for "tit2 cap4" was produced, either by its own alteration or indirectly by the alteration of one or more components in its tree. A more common implementation would be to produce new versions for the other children of "tit2" on 1993-09-14 and then produce a 1993-09-14 version for "tit2", which would be a *composition*¹³ of its children's versions on the same date - similar to the formation of the "tit2" text on the original date. This would imply that, for each date on which the text of the norm was altered, a complete version of the norm would be produced from the composition of the complete versions of each immediate child of its articulation.

However, the other children of "tit2" were not altered on 1993-09-14. Our proposal, therefore, is not to produce versions for the unaltered children (for dates other than the original) and instead reuse the most recent versions of these children. Instead of composition, the 1993-09-14 version for "tit2" would be an *aggregation*¹⁴ of the most recent versions of the children.

We can verify the formation of these version aggregations in Figure 11, where new versions of the component "tit2" were produced as a consequence of new versions of one or more of its children on each date, reusing unchanged versions of its children where appropriate.

Thus, in the construction of the graph, besides each version being existentially linked to its component,

¹³In UML terms, Composition represents a strong whole-part relationship where the "part" belongs exclusively to one "whole" and shares its lifecycle; if the whole is deleted, the part is too. In our initial comparison, this would mean a parent version strictly owns child versions from the exact same date.

¹⁴Aggregation, in UML, represents a weaker whole-part relationship where the "part" can exist independently of the "whole" and can potentially be shared by multiple wholes. In our proposed model, this reflects how a child version (the "part") can be reused by multiple parent versions (the "wholes") across different times without its existence depending on any single one.

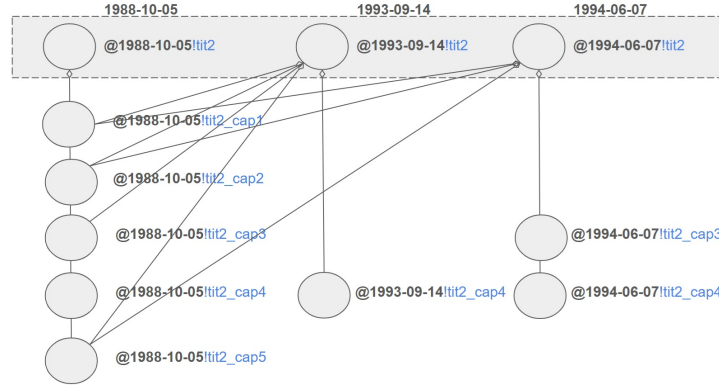


Figure 11: New versions of the component "tit2" derived from new versions of some of its children

this work proposes that each version, when it is not a leaf, be a *agregation* with its child versions. In other words, a child version does not "belong" exclusively to a single parent version, but rather contributes to the formation of multiple parent versions at different moments in time.

When a version of the child component C_x does not exist for a date T_n , the most recent version of C_x , prior to T_n , will be used in the aggregation, as exemplified by V1 of C3, which is being used in the formation of version V2 of C1 (examples of this situation also appear in Figure 11). In this case, Version V1 of component C3 contributes to versions V1 and V2 of component C1, demonstrating that a child version can participate in multiple parent versions in different temporal contexts. The orange arrows indicate the aggregation relationship (Figure 12).

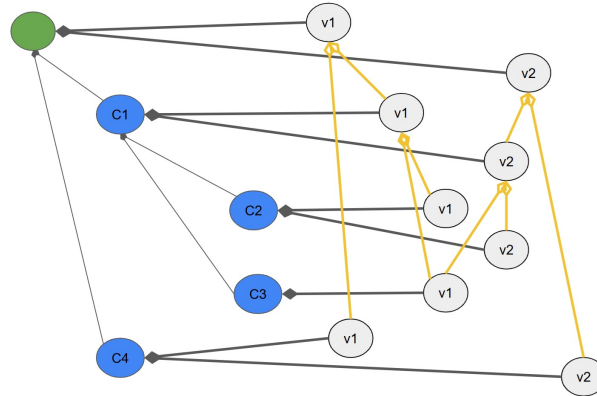


Figure 12: Diagram of aggregation relationships between versions, illustrating how child versions can contribute to multiple parent versions at different times and how earlier versions are reused in the absence of subsequent changes.

This representation of versions as *aggregation* ensures uniqueness in the representation of a version. In a traditional approach, there is a complete version of the articulation tree for each change date. That is, for each component of the tree, there is a version for the date, even if the content of that component was not changed on that date. In the proposed approach, there is a version for the component on the original date and on each date its content was actually changed, thus eliminating redundant and unnecessary copies of the version, contributing to a non-ambiguous and economical representation.

In the context of RAG, it allows the system to retrieve not only the specific version of a component, but also the context of its parent and child versions, providing a more holistic and precise understanding of the

norm at a given moment, facilitating more complex queries about the temporal evolution of norms [2].

3.3 Additional Context for Versions

This work also proposes that, for each *Version* inserted into the graph, information describing its validity status and, when terminated, the type of action that promoted its termination (“*revogação*” (repeal), “*alteração de texto*” (text change), “*supressão*” (suppression), “*derrubada de veto*” (veto override), among other examples) and the responsible norm (the legal norm, and, ideally, the specific component of that norm, that promoted or commanded the action) be added, when available, as additional context of the version.

This information of a version will be registered by the property *Contexto Adicional* (Additional Context) of the *Version* entity in the graph. If new alteration actions occur between subsequent and prior norms, this status can be updated over time.

For example, consider the caput of Art. 6° (presented in Figure 5) of the Brazilian Federal Constitution of 1988, signed and published on 1988-10-05. Thus, at the time of the Constitution’s publication, the Additional Context property of its original version (1988-10-05) will contain information about the start of its validity:

"Essa versão tem vigência de 1988-10-05 até hoje."
("This version is valid from 1988-10-05 until today.")

The *version* of a component is created using the **signature date** of its norm (when original) or the signature date of the altering norm, when produced by an action. On the other hand, the **validity** of the version starts from the **publication date** (by default in Brazil)¹⁵ of the norm (when original) or the publication date of the altering norm, when produced by an action – and the previous version of the component, when exists, has its validity terminated using the date before this publication date.

For example, with the publication, on 2000-02-15, of “Emenda Constitucional nº 26 de 14/02/2000” (Brazilian Constitutional Amendment No. 26), signed on 2000-02-14, an action was commanded, through the caput of its Art. 1°, to alter the wording of the caput provision of Art. 6° of the Brazilian Federal Constitution of 1988 – the action is represented in the graph¹⁶ by the *Action* entity (in brown in Figure 13). This action terminated the validity of the 1988-10-05 version and produced a new version, dated 2000-02-14 (with the new text), for the altered component.

For a more precise representation in the Graph, specialized entity types, derived from *Action*, can be created for each type of action between norms found in the treated legal system. For example, in the Brazilian case, types like “Alteração de Texto” (Text Change Action), “Renumeração” (Renumbering Action), “Revogação” (Repeal Action) etc. Thus, the entity representing the action could be included in the graph using the specialized type.

The new version, dated 2000-02-14 (signature date of the Amendment), of the caput of Art. 6° of the Brazilian Federal Constitution of 1988, will then have as the content of the Additional Context, at the time of the Amendment’s publication:

"Essa versão tem vigência de 2000-02-15 até hoje."
("This version is valid from 2000-02-15 until today.")

Furthermore, the content of the Additional Context of the previous version (dated 1988-10-05) of the caput of Art. 6° of the Brazilian Federal Constitution of 1988 will be updated, with a complement textually describing the reason for and the entity responsible for the termination of its validity:

¹⁵Defining the start of validity for a legal norm and its components is a complex issue, due to a wide variety of possibilities. Therefore, it will not be the focus of this article and will be explored in future work.

¹⁶The modeling of the “Action presented in Figure 13 was simplified for didactic purposes. It can be improved using ontology techniques, such as those proposed in UFO (Unified Foundational Ontology), found at <https://nemo.inf.ufes.br/projetos/ufo/>. Accessed on 22.03.2025.

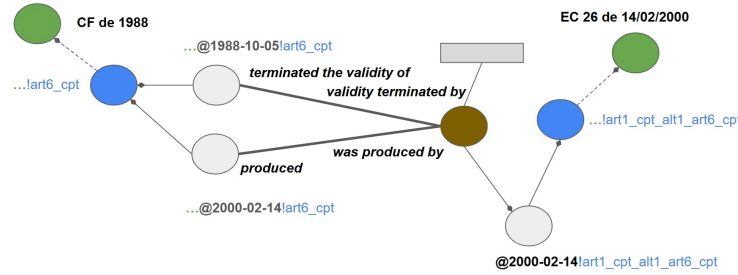


Figure 13: Representation in the knowledge graph of the text change action

"Essa versão tem vigência de 1988-10-05 a 2000-02-14. Sua vigência foi encerrada pela nova redação de texto dada a esse dispositivo pelo caput do Art. 1º da Emenda Constitucional nº 26 de 14/02/2000."
 ("This version was valid from 1988-10-05 to 2000-02-14. Its validity was terminated by the new text wording given to this component by the caput of Art. 1º of Constitutional Amendment No. 26")

The additional contexts can be persisted directly as textual metadata (property) of the version or be generated automatically from the actions (and from **start-of-validity** and **end-of-validity** information) inserted in the graph.

Either way, the content of the Additional Context should not be used in the creation of the Text Unit embedding of its version, so as not to add noise to its semantic understanding.

The Additional Context is important to enrich the response generated by the LLM in the RAG process. When the RAG system retrieves a specific *Version* as relevant context, the Additional Context associated with this version can be sent along with the Text Unit to the LLM, providing valuable information about the history and validity of the component, allowing for more complete and contextually rich responses – in more advanced implementations, the additional context would be sent to the LLM only when it is sensed that it is necessary for generating the response. This will provide context for the LLM to display, for example, besides the texts in the timeline of a component, also information about the validity of each text and, in case of termination, by whom and for what reason its validity was terminated.

3.4 Alteration Actions as Text Units

As previously presented, a subsequent norm can mandate an amendment to the wording of a provision of a prior norm, a repeal of a provision, or of a prior norm in its entirety, etc. An additional proposal, therefore, is to create a Text Unit representing each alteration action between legal norms/components. Considering the action illustrated in Figure 13, the content of the new Text Unit will be:

"A Emenda Constitucional nº 26, de 14 de fevereiro de 2000, por meio do caput do seu Art. 1º, deu uma nova redação ao caput do Art. 6º da Constituição Federal de 1988. Essa alteração encerrou em 2000-02-14 a vigência da versão original desse dispositivo (de 1988-10-05) e instituiu uma nova versão com vigência a partir de 2000-02-15, cujo texto passou a ser: 'São direitos sociais a educação, a saúde, o trabalho, a moradia, o lazer, a segurança, a previdência social, a proteção à maternidade e à infância, a assistência aos desamparados, na forma desta Constituição.'
 ("Constitutional Amendment no. 26, of February 14, 2000, through the caput of its Art. 1º, provided a new wording for the caput of Art. 6º of the Brazilian Federal Constitution of 1988. This alteration terminated on 2000-02-14 the validity of the original version of this provision (from 1988-10-05) and established a new version effective from 2000-02-15, whose text became: 'Social rights include education, health, work, housing, leisure, security, social security, protection of motherhood and childhood, and assistance to the destitute, in the manner prescribed by this Constitution.'")

Although this alteration command already appears in the text of the caput of Art. 1° of Constitutional Amendment no. 26 (amending component/norm), as shown in Figure 14, generating a Text Unit for the *Action* entity structures and completes the information about the action/command, making explicit, for example, the validity periods of the text versions.

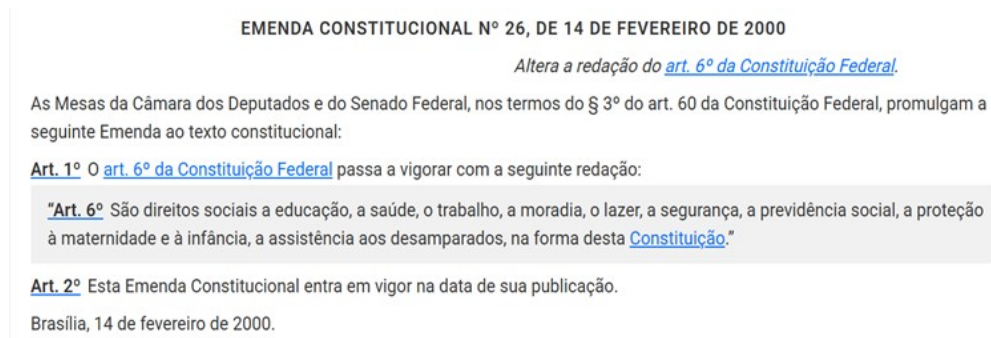


Figure 14: Text of the Emenda Constitucional nº 26 de 14/02/2000 (Constitutional Amendment no. 26)

The generated Text Unit is then associated with the *Action* entity in the graph. And it receives its own embedding, allowing, besides *Version* Text Units embeddings, for more sophisticated analyses of legislative evolution, facilitates the traceability of normative alterations, and improves the understanding of the life cycle of legal norms in the knowledge graph in the context of RAG [14].

3.5 Metadata as Text Units

Besides the texts of legal norms, other sources of information can be used to feed the knowledge graph. One possibility would be to generate Text Units with the metadata¹⁷, in textual format, to associate them with the *Norm* and *Component* entities, similar to the original proposal of entity summaries from Graph RAG – the difference here is that this information, generally, is not contained within the text of the norm itself; therefore, these summaries could not be extracted from the text chunks, as in the original proposal.

As example, for the *Norm* entity representing the legal norm “Lei nº 10.406 de 10/01/2002” (Law no. 10,406 of 2002-01-10), a Text Unit would be generated textually describing the entity’s properties¹⁸:

"São apresentados a seguir os metadados gerais da norma jurídica federal 'Lei nº 10.406 de 10/01/2002': é do tipo 'Lei Ordinária'; está no nível 'Legal' do ordenamento jurídico brasileiro; possui como urn: 'urn:lex:br:federal:lei:2002-01-10;10406'; possui como URL principal (uri): '<https://normas.leg.br/?urn=urn:lex:br:federal:lei:2002-01-10;10406>'; possui como URLs alternativas: '<http://www.lexml.gov.br/urn/urn:lex:br:federal:lei:2002-01-10;10406>' e '<http://legis.senado.leg.br/legislacao/DetalhaSigen.action?id=552282>'; foi apreciada e aprovada pelo 'Congresso Nacional'; foi assinada em Brasília, 10 de janeiro de 2002; 181º da Independência e 114º da República; foi assinada por FERNANDO HENRIQUE CARDOSO, Aloysio Nunes Ferreira Filho; foi oficialmente publicada pela 'Imprensa Nacional', tendo como data de publicação original e oficial: 11/01/2002; tem como nomes alternativos: 'LEI-10406-2002-01-10' e 'Código Civil (2002) (CC)'; tem como ementa: 'Institui o Código Civil'."

("The general metadata for the federal legal norm Law no. 10,406 of 2002-01-10 are presented below:

¹⁷Metadata is data about data. In this context, structured or textual information describing the properties and characteristics of legal norms, such as publication date, authorship, type of norm, relationships with organizations, etc.

¹⁸This Text Unit was generated from the structured metadata registered in the knowledge graph of the "Sistema de Gestão de Normas Jurídicas (Sigen)" (Legal Norm Management System) of the Federal Senate of Brazil. This metadata can also be viewed publicly, in structured form, through the Normas.leg.br portal, as in the example: <https://normas.leg.br/?urn=urn:lex:br:federal:lei:2002-01-10;10406>. Accessed on 2025-03-22.

it is of the type 'Ordinary Law'; it is at the 'Legal' level of the Brazilian legal system; its URN is: 'urn:lex:br:federal:lei:2002-01-10;10406'; its main URL (URI) is: '<https://normas.leg.br/?urn=urn:lex:br:federal:lei:2002-01-10;10406>'; its alternative URLs are: '<http://www.lexml.gov.br/urn/urn:lex:br:federal:lei:2002-01-10;10406>' and '<http://legis.senado.leg.br/legislacao/DetalhaSigen.action?id=552282>'; it was reviewed and approved by the 'National Congress'; it was signed in Brasília, on January 10, 2002; 181st year of Independence and 114th year of the Republic; it was signed by FERNANDO HENRIQUE CARDOSO, Aloysio Nunes Ferreira Filho; it was officially published by the 'National Press', with the original and official publication date being: 2002-01-11; it has alternative names: 'LEI-10406-2002-01-10' and 'Civil Code (2002) (CC)'; its summary is: 'Institutes the Civil Code'."

Informative relationships between legal norms, such as regulation, succession, correlation, etc., do not promote actions that alter the text of a norm, thus not generating temporal versions. Nevertheless, they are important in the context of semantic retrieval of legal information. Therefore, we could also generate Text Units with the textual descriptions of these relationships.

Considering the legal norm 'CONSTITUIÇÃO DO BRASIL (1967)' (Constitution of Brazil (1967)), the following are the Text Units¹⁹ for its relationships presented in the graph of Figure 15.

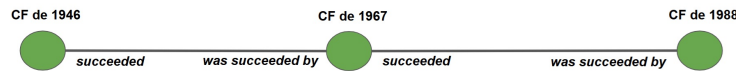


Figure 15: Example representation of inter-normative relationships (succession) between different normative acts in the knowledge graph.

A norma jurídica 'CONSTITUIÇÃO DO BRASIL (1967)' sucedeu a norma jurídica 'CONSTITUIÇÃO DOS ESTADOS UNIDOS DO BRASIL (1946)'

(The legal norm 'Constitution of Brazil (1967)' succeeded the legal norm 'Constitution of the United States of Brazil (1946)').

A norma jurídica 'Constituição da República Federativa do Brasil' sucedeu a norma jurídica 'CONSTITUIÇÃO DO BRASIL (1967)'

(The legal norm 'Constitution of the Federative Republic of Brazil' succeeded the legal norm 'Constitution of Brazil (1967)')

This approach aligns with the concept of “Multi-aspect embeddings” [19], which proposes using multiple vectors to represent information at different levels of detail or different aspects. Instead of generating a single, complete Text Unit for an entity encompassing all its properties and relationships, this work proposes generating one Text Unit for the entity’s intrinsic properties (general metadata), as exemplified above, and a separate Text Unit for each of its relationships.

Thus, instead of having just one vector/point in the vector space representing an entity, we will have multiple vectors/points associated with it. Each vector will capture/encode a different aspect of the information related to the entity (potentially viewed as a different “lens” through which the entity is represented), thus offering a more contextualized and granular view of the entity, which can lead to significant improvements in semantic retrieval by proximity.

It is also proposed to generate Text Units for *Component* entities. For example:

"O dispositivo 'Art. 5º, caput, XII –' da 'Constituição da República Federativa do Brasil' é correlato da norma jurídica 'Lei nº 14.063 de 23/09/2020'."

¹⁹It is not necessary to generate a Text Unit for the other side of the relation, since an LLM and a RAG retrieval already semantically understand this reversal ("succeeded" vs "was succeeded by", for example).

("The provision 'Art. 5º, caput, XII -' of the 'Constitution of the Federative Republic of Brazil' is correlated with the legal norm 'Law no. 14,063 of 2020-09-23'.")

These *metadata Text Units* about the norm, or component, can be generated externally (from other information sources) and inserted directly into the graph, or be generated from the properties and/or relationships of the entity in the graph²⁰ - although the original metadata might be structured in the graph, converting them to textual format allows their uniform integration and processing within the Graph RAG workflow.

The *metadata Text Unit* for a norm or component will have its own embedding generated and will also be subject to semantic retrieval, like the content Text Units derived from versions and alteration actions. Consequently, vector search will be able to retrieve information about the metadata (properties and relationships) of a legal norm (or a component), even though this information is generally not present in the text of the norm itself.

We can see these Text Units derived from metadata, additional to those derived from versions of the norms/components, in Figure 16. The insertion of this type of Text Unit brings additional information to the Knowledge Graph and, consequently, to the vector database of contexts.

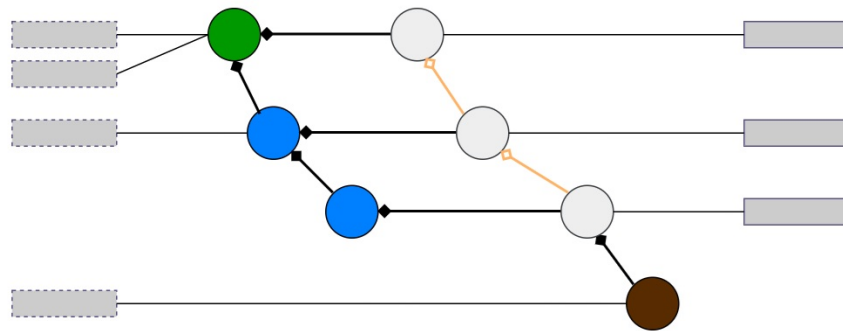


Figure 16: Knowledge graph illustrating the Text Units derived from both versions of the norms/components and their metadata.

3.6 Hierarchical Grouping as Communities

The grouping components of legal norms function in practice as hierarchical levels for thematic grouping, chosen as the systematizing criterion for the articulation. As observed in the example of Figure 4, Art. 5º (Article 5) is grouped under “CAPÍTULO I – DOS DIREITOS E DEVERES INDIVIDUAIS E COLETIVOS” (CHAPTER I – OF INDIVIDUAL AND COLLECTIVE RIGHTS AND DUTIES), which in turn is grouped under “TÍTULO II - DOS DIREITOS E GARANTIAS FUNDAMENTAIS” (TITLE II - OF FUNDAMENTAL RIGHTS AND GUARANTEES). These groupings then ascend in level until reaching the *Norm* (as a Whole), representing natural hierarchical and thematic structures within legal norms. This directly aligns with the concept of communities in a graph proposed by the Graph RAG approach – in this approach, “communities” refer to groups of nodes in the graph that are more densely interconnected among themselves than with the rest of the graph, representing thematic or conceptual clusters. Therefore, the hierarchical structure of legal norms can be viewed as an intrinsic form of organization into thematic communities.

²⁰As these properties (publication date, alternative names, level in the legal system, etc.) and relationships (correlation, succession, etc.) generally cannot be directly derived from the texts of the legal norms, they must be populated in the graph either manually or through semi-automatic extraction processes using other information sources.

Legal norms, in general, are classified into one or more themes²¹, which can be considered intertextual, or inter-norm, aggregators, transversally gathering norms on cohesive subjects, as exemplified in Figure 17. In this figure, the *Norm* (as a whole) are represented by circles in green, and the *Themes*, in orange.

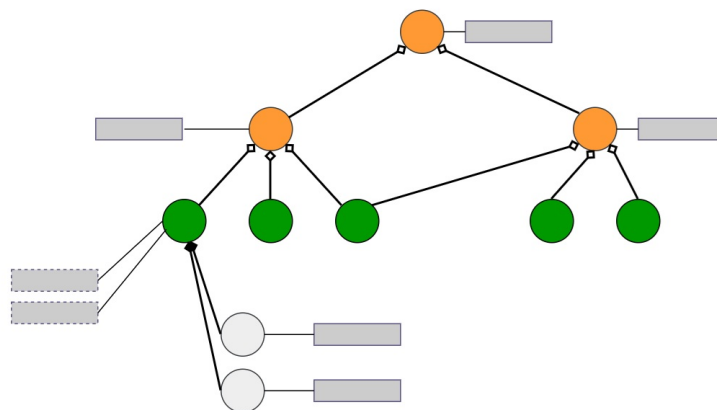


Figure 17: Diagram illustrating the inter-norm aggregation by legal themes (orange), represented as higher-level communities that group "Norms" (green) in the knowledge graph.

Components of legal norms could also be classified under one or more themes, as demonstrated in Figure 18. This allows broadening the classification of the components beyond their groupers within the hierarchical structure of the legal norm.

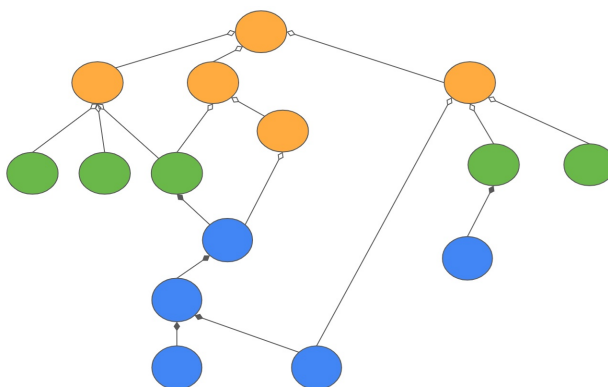


Figure 18: Diagram illustrating associations between components and legal themes.

The themes would then be inserted as Entities/Nodes *Theme* in the Knowledge Graph, each having an associated Text Unit with its description, which would be produced by a human domain expert, ensuring the curation of the produced texts.

Another approach could be to use an LLM to generate the theme descriptions, producing summaries from the descriptions of the sub-themes or, in the case of leaf themes, from the summaries of the associated norms - it would require intensive processing and high costs (considering that a theme can cover hundreds of norms). The LLM would act as an assistant in a semi-automatic generation, where these descriptions would be reviewed and potentially updated by the human domain expert - the summary's effectiveness would

²¹Brazilian federal norms are classified according to the "Sistema de Classificação de Normas Jurídicas do Congresso Nacional" (System for Classification of Legal Norms of the National Congress), with examples of themes including: "Children and Adolescents", "Extraordinary Credit", "Indigenous Population", "Basic Sanitation", "General Social Security System", etc.

heavily depend on the quality of the algorithms and models used.

3.7 Retrieval considering Structure

Another advantage of using a graph to represent structural entities, reflecting the hierarchical structure of legal texts, is that the user could choose the scope of their query (level of aggregation), for example, by defining that they wish to query about a specific theme (which would encompass all its sub-themes and associated norms, with their components and versions), or about a specific title or article of a norm (which would encompass all its subdivisions and versions), or about a specific version of a norm or a component (which would encompass its child versions), as presented in Figure 19.

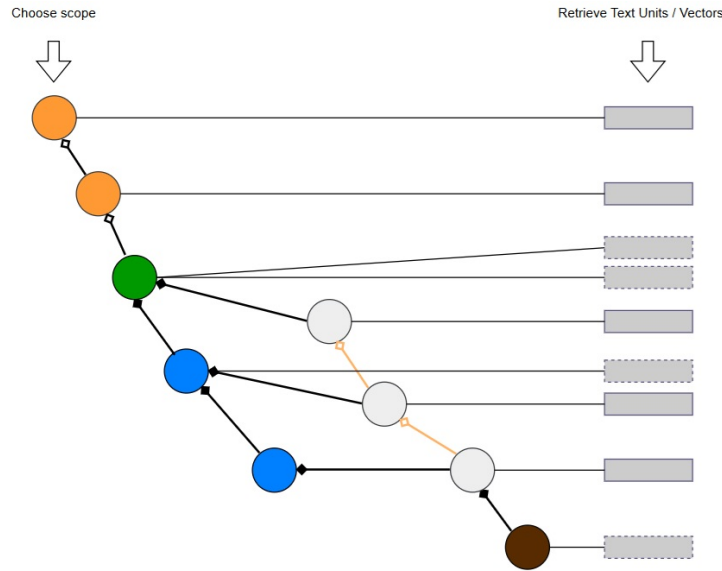


Figure 19: Diagram illustrating the possibility of retrieving Text Units considering the hierarchical structure of the knowledge graph

The choice defines a filter on the hierarchical index of the Text Units (and vectors/embeddings) to be retrieved, making the context retrieval for the response more focused. The hierarchical structure of the graph allows for more precise semantic navigation between different levels of detail. This “*semantic navigation*” manifests in the user’s ability to direct the search to a specific level of the legal hierarchy (theme, title, chapter, article, etc.), filtering the search space and focusing retrieval on the Text Units most relevant to their interest at that level of detail. Instead of a generic search across the entire graph, the user can explore legal knowledge in a more structured and hierarchy-guided manner.

4 Conclusion and Future Work

Aiming to enhance the analysis and comprehension of legal norms in the era of Artificial Intelligence, this article presented a detailed proposal for the construction of a legal knowledge graph, enriched with contextually relevant Text Units. The central objective was to expand the scope of queries regarding legal norms, allowing Artificial Intelligence systems to address questions ranging from literal text retrieval to complex inquiries requiring an understanding of the hierarchical structure, the intrinsic semantics, the intra- and inter-relations and the temporal evolution of law.

Incorporating such structural and temporal knowledge is expected to enhance the precision of legal AI responses, thereby reducing contextual errors and the risk of citing incorrect versions of the law.

In summary, the main contributions of this work include:

- A detailed proposal for building a legal knowledge graph, integrating the hierarchical structure, temporal versioning, and cumulative and contextually enriched Text Units, with an initial focus on structural entities.
- The modeling of *Versions as Aggregations*, reflecting hierarchical interdependence and the propagation of changes over time, allowing for an accurate and efficient representation of normative evolution.
- The suggestion to incorporate *Additional Context* into versions, enriching the graph with descriptive metadata about the history and lifecycle of norms, providing additional context during response generation by the LLM.
- The proposal to generate specific Text Units for *Alteration Actions* (e.g., repeal, text amendment), making the norm-modifying events themselves and their effects semantically retrievable.
- The generation of Text Units from *structured Metadata* (intrinsic properties and relationships such as succession or correlation), allowing non-explicitly textual information in norms to be included in vector search and aligning with concepts like multi-aspect embeddings.
- The perspective of *Hierarchical Grouping as Communities*, interpreting the hierarchical structure and legal themes (curated by specialists) as thematic communities within the graph, contributing to more targeted and semantic information retrieval strategies.
- The proposal of *Retrieval considering Structure*, enabling users to define the search scope and semantically navigate the legal hierarchy (whether by theme, title, article, version, etc.), refining the precision and relevance of responses.
- A Graph RAG approach that seeks the lowest possible cost for graph creation by prioritizing pre-existing or curated hierarchical and thematic structures, avoiding computationally intensive techniques such as automatic community detection and the construction of complex summaries by LLMs for this purpose.

The main innovation lies in the adaptation and expansion of Graph RAG to the legal field, with a particular focus on structural entities, considered essential for identifying robust legal grounds. Semantic segmentation guided by the hierarchical structure, combined with the representation of cross-references, versioning and grouping into thematic communities, provides a natural and rich contextual basis for legal RAG strategies.

The proposal to use the hierarchy of communities and themes designed by domain experts reflects the original intent of legislators and ensures informational curation, an essential aspect for the reliability of AI systems in Law, in addition to minimizing the need for complex and costly graph construction processes.

For future work, several research possibilities open up:

- It is fundamental to empirically validate the effectiveness of the proposed Graph RAG approach through experiments with legal norm datasets and rigorous evaluations of the quality and precision of the responses generated by LLMs.
- Although the proposal of this work significantly advances the representation of normative complexity, it primarily focused on modeling the hierarchical structure and temporality. The deep extraction and semantic integration of Content Entities (people, organizations, legal concepts etc.) and their complex interrelationships [5] remain a challenge reserved for future investigations.

- Investigating the use of Artificial Intelligence techniques, such as language models and clustering algorithms, to support human curation in detecting thematic communities and suggesting hierarchical structures represents another promising direction.
- Exploring the practical application of legal Graph RAG in legal decision support systems, legislative impact analysis tools, and platforms for facilitated access to legal information for legal professionals and citizens can demonstrate the concrete value and feasibility of the approach in real-world and diverse scenarios.
- Integrating Bigraphs [18] with the proposed Graph RAG model could further enhance the representation of spatio-temporal relationships and enable dynamic memory operations, such as automatic version consolidation or the identification of normative conflicts via semantic link analysis.
- Investigating the applicability and potential adaptation of the proposed Graph RAG approach beyond legal norms to other fundamental types of legal documents, such as jurisprudence (case law) and contracts. While the current focus leverages the predefined hierarchy of norms, exploring how this framework could be extended or modified to handle the different structural conventions (e.g., case structure, contract clauses) and temporal dynamics (e.g., precedential relationships, contract amendments) inherent in these other domains would be a valuable direction. This analysis would help clarify the approach’s scope of generality and its limitations, further contextualizing its contribution within the broader field of Legal AI.

Ultimately, by integrating structure, cross-references, temporality, and semantics into a legal knowledge graph, this work offers a step towards more intelligent, transparent, and effective Artificial Intelligence systems for the complex and fundamental domain of Law. By facilitating precise information retrieval and making legislative analysis more accessible through a navigable and contextualized representation, this approach has the potential to democratize access to legal information and empower professionals and citizens in an increasingly regulated world.

References

- [1] Vinx L. Hans Kelsen’s pure theory of law: legality and legitimacy. Oxford: Oxford University Press; 2007.
- [2] Chen Z, Li D, Zhao X, Hu B, Zhang M. Temporal knowledge question answering via abstract reasoning induction. arXiv preprint [arXiv:2311.09149v2](https://arxiv.org/abs/2311.09149v2) [Preprint]. 2023 [cited 2024 Apr 23]. Available from: <https://arxiv.org/abs/2311.09149v2>
- [3] Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *Adv Neural Inf Process Syst*. 2020;33:9459-74.
- [4] Edge D, Trinh H, Cheng N, Bradley J, Chao A, Mody A, et al. From local to global: A graph rag approach to query-focused summarization. arXiv preprint [arXiv:2404.16130](https://arxiv.org/abs/2404.16130) [Preprint]. 2024 [cited 2024 Apr 23]. Available from: <https://arxiv.org/abs/2404.16130>
- [5] Li J, Yang G, Xu C, Zhang Y, Li H, Huang Z, et al. Construction of legal knowledge graph based on knowledge-enhanced large language models. *Information*. 2024;15(11):666.
- [6] Kejriwal M. Knowledge graphs: A practical review of the research landscape. *Information*. 2022;13(4):161.

- [7] Mikolov T, Chen K, Corrado G, Dean J. Efficient estimation of word representations in vector space. arXiv preprint [arXiv:1301.3781](https://arxiv.org/abs/1301.3781) [Preprint]. 2013 [cited 2024 Apr 23]. Available from: <https://arxiv.org/abs/1301.3781>
- [8] Dang DV, Le NT, Nguyen NT, Takasu A. Information retrieval from legal documents with ontology and graph embeddings approach. In: Proceedings of the International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems; 2023. p. 300-12.
- [9] Brazil. Complementary Law No. 95 of February 26, 1998. Provides for the consolidation, codification, and drafting of laws. Diário Oficial da União. 1998 Feb 27 [cited 2024 Apr 23]. Available from: <https://normas.leg.br/?urn=urn:lex:br:federal:lei.complementar:1998-02-26;95>
- [10] Roddick JF, Spiliopoulou M. A survey of temporal knowledge discovery paradigms and methods. IEEE Trans Knowl Data Eng. 2002;14(4):750-67.
- [11] He X, Tian Y, Sun Y, Chawla N, Laurent T, LeCun Y, et al. G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. Adv Neural Inf Process Syst. 2023;36:132876-907.
- [12] Lima JAO. Unlocking legal knowledge with multi-layered embedding-based retrieval. arXiv preprint [arXiv:2411.07739](https://arxiv.org/abs/2411.07739) [Preprint]. 2024 [cited 2024 Apr 23]. Available from: <https://arxiv.org/abs/2411.07739>
- [13] Libal T. Legal linguistic templates and the tension between legal knowledge representation and reasoning. Front Artif Intell. 2023;6:1136263.
- [14] Tang Y, Gu Y, Qu S, Cheng X. CaseGNN: Graph neural networks for legal case retrieval with text-attributed graphs. In: Proceedings of the European Conference on Information Retrieval; 2024. p. 80-95.
- [15] Reddy RG, Garg D, Oh A, Kumar S, Muresan S, Richardson F, et al. AGRaME: Any-granularity ranking with multi-vector embeddings. arXiv preprint [arXiv:2405.15028](https://arxiv.org/abs/2405.15028) [Preprint]. 2024 [cited 2024 Apr 23]. Available from: <https://arxiv.org/abs/2405.15028>
- [16] Chalkidis I, Fergadiotis M, Malakasiotis P, Aletras N, Androutsopoulos I. LEGAL-BERT: The muppets straight out of law school. arXiv preprint [arXiv:2010.02559](https://arxiv.org/abs/2010.02559) [Preprint]. 2020 [cited 2024 Apr 23]. Available from: <https://arxiv.org/abs/2010.02559>
- [17] Salton G. Automatic Text Processing: The Transformation, Analysis, and Retrieval of Information by Computer. Boston: Addison-Wesley; 1989.
- [18] Pavlyshyn V. Bigraphs for AI agent memory [Internet]. Medium. 2023 [cited 2024 Apr 27]. Available from: <https://volodymyrpavlyshyn.medium.com/bigraphs-for-ai-agent-memory-3bdae9809f4a>
- [19] Park C, Yang C, Zhu Q, Kim D, Yu H, Han J. Unsupervised differentiable multi-aspect network embedding. In: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '20); 2020. p. 1435-1445.